

EXPERIMENT 7

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Experiment 6 – Understand Database Normalisation

Aim: To understand NF1, NF2, NF3, BCNF and problem solving using Relations and Function dependency.

Find Highest Normalisation form of the given Relations.

Problem 1: R(ABCDE)

CE→D

D→B

C→A

Sol:

Candidate key is CE

Relation not in BCNF as non-super key is determining non-prime key. D→B

Relation not in 3NF as non-prime attribute determining non-prime attribute

Relation not in 2NF as subset of Candidate key is determining a non-prime attribute.

Hence Relation is in 1NF

Problem 2: R(ABCDEF)

AB→C

DC→AE

E→F

Sol: Candidate key is ABD

Relation is in 1NF as subset of candidate is determining non-prime attribute

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Problem 3: R(ABCDEFGHI)

AB→C

BD→EF

AD→GH

A→I

Sol:

Candidate key is ABD

Relation is in 1NF.

Subset of candidate key is determining non-prime attribute

Problem 4: R(ABCDE)

AB→CD

D→A

BC→DE

Sol:

Candidate Key is BC

Relation is in 2NF as partial dependency is removed and non-super key is determining the attribute

Problem 5: R(ABCDE)

BC → ADE

D → B

Sol:

Candidate Key is BC , CD

Relation is in 3NF, non-super key is determining the attribute.

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Problem 6: R(ABC)

A→B

B→C

C→A

Sol:

Candidate key is A,B,C

Relation is in 3NF, prime attributes determining prime attribute but non-super key determining the attribute

Problem 7: R(F1, F2, F3, F4, F5)

F1→F3

F2→F4

(F1,F2)→F5

Sol: Candidate key is (F1 F2)

Relation is in 1NF, subset of candidate key determining attribute.

Problem 8: R(ABCDEFGHI)

AB → C

AD → GH

BD → EF

A → I

H → J

Sol:

Candidate key is ABD

Relation is in 1NF as subset of candidate key determining attribute.

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Problem 9: R(ABCDE)

A $\rightarrow$ B

B $\rightarrow$ E

C $\rightarrow$ D

Sol:

Candidate key is AC

Relation is in 1NF, subset of candidate key is determining the attribute.

Learning Outcome

After completing this experiment, the learner will be able to:

1. Understand Functional Dependencies.
2. Understand Database Normalisation forms.
3. Understand NF1, NF2, NF3, BCNF.